

Giovanni Varricchione

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Work Experience

Universiteit Utrecht

Mar 2021 - Feb 2026

PhD in Neurosymbolic Reinforcement Learning

- Supervised by Prof. Dr. Mehdi Dastani and Prof. Dr. Brian Logan
- Leveraged logic and formal methods to develop safe, transparent, and efficient reinforcement learning approaches
- Modified sub-symbolic deep RL algorithms by augmenting them with techniques based on temporal logics and symbolic AI
- Led and took part in research groups with foreign researchers
- Published and presented research at top AI venues such as AAAI, KR, and JAIR
- Various teaching duties: several supervisions of bachelor's and master's students, supervision of tutorials and project groups, lecturing and course organization

Education

Universiteit van Amsterdam

Sep 2019 - May 2021

MSc in Logic (cum laude)

- Thesis: Complexity of Locally Fair Allocations on Graphs
- GPA: 8.98/10, thesis grade: 9/10
- Focus on computability, reasoning, computational social choice and game theory

Università degli Studi di Roma "La Sapienza"

Sep 2018 - Aug 2019

MSc in Computer Science (transferred)

- Classes completed: Models of Computation, Machine Learning, Mathematical Logic for Computer Science, Biometric Systems, Natural Language Processing

Università degli Studi di Roma "La Sapienza"

Sep 2015 - Oct 2018

BSc in Computer Science (cum laude)

- Thesis: Automatic Versions of Combinatorics Theorems
- Final grade: 110/110

Publications

Journal

- [J-1] GV, Natasha Alechina, Mehdi Dastani, and Brian Logan. Synthesising reward machines for cooperative multi-agent reinforcement learning. *Journal of Artificial Intelligence Research*, 85(38), 2026

Conference Proceedings

- [C-1] Aronis Panos, Mehdi Dastani, Roxana Rădulescu, and GV. Leveraging reward machines for efficient multi-objective reinforcement learning. In *Proceedings of the 18th Adaptive and Learning Agents Workshop (ALA)*, 2026
- [C-2] GV, Toryn Q. Klassen, Natasha Alechina, Mehdi Dastani, Brian Logan, and Sheila A. McIlraith. Pushdown reward machines for reinforcement learning. In *Proceedings of the 22nd International Conference on Principles of Knowledge Representation and Reasoning (KR)*, 2025
- [C-3] Luca Barbaro, GV, Marco Montali, and Claudio Di Ciccio. From sound workflow nets to LTLf declarative specifications by casting three spells. In *Proceedings of the BPM Forum at BPM 2025*, 2025
- [C-4] GV, Natasha Alechina, Mehdi Dastani, and Brian Logan. Maximally permissive reward machines. In *Proceedings of the 27th European Conference on Artificial Intelligence (ECAI)*, 2024
- [C-5] Damiano Fornasiere, Johannes Marti, and GV. Frame definability in conditional logic. In *Proceedings of the Conference on Advancements in Modal Logic (AiML)*, 2024
- [C-6] GV, Natasha Alechina, Mehdi Dastani, Giuseppe De Giacomo, Brian Logan, and Giuseppe Perelli. Pure-past action masking. In *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI); Safe, Robust and Responsible AI track*, 2024
- [C-7] GV, Natasha Alechina, Mehdi Dastani, and Brian Logan. Synthesising reward machines for cooperative multi-agent reinforcement learning. In *Proceedings of the 20th European Conference on Multi-Agent Systems (EUMAS)*, 2023

Workshops and Meetings (Peer Reviewed)

- [W-1] GV, Natasha Alechina, Mehdi Dastani, Giuseppe De Giacomo, Brian Logan, and Giuseppe Perelli. Pure-past action masking. BNAIC/BeNeLearn 2024; Type B (Encores), 2024
- [W-2] GV, Natasha Alechina, Giuseppe Dastani, and Brian Logan. Maximally permissive reward machines. BNAIC/BeNeLearn 2024; Type B (Encores), 2024
- [W-3] GV, Natasha Alechina, Mehdi Dastani, Giuseppe De Giacomo, Brian Logan, and Giuseppe Perelli. Pure-past action masking. Logic Colloquium 2024
- [W-4] GV, Natasha Alechina, Mehdi Dastani, Giuseppe De Giacomo, Brian Logan, and Giuseppe Perelli. Preemptive restraining bolts. Bridging the Gap Between AI Planning and Reinforcement Learning (PRL @ ICAPS 2023) – Workshop at ICAPS 2023
- [W-5] GV, Natasha Alechina, Mehdi Dastani, and Brian Logan. Synthesising reward machines for cooperative multi-agent reinforcement learning. International Workshop on Logical Aspects in Multi-Agent Systems and Strategic Reasoning, LAMAS&SR'22, 2022

Teaching experience and activities

Logics for Safe AI

MSc AI, Lecturer, 23/24-25/26

Designed and given lectures for 20% of the course (2 weeks out of 9), designed and graded one out of four coursework assignments, designed and graded exam questions. The lectures were about

logic-based approaches for (safe) reinforcement learning. Adapted material, especially for lectures, every year by using student feedback. Approximately 30/40 students took the course each year.

Logica voor Informatica

BSc Informatica Year 1, Tutor, 22/23-24/25

Led tutorial groups (~20 students) to cover exercises about the lectures' material. In 23/24 and 24/25 also led a (sub)group of TAs, and supervised each group during each tutorial session. Gave a lecture in 24/25 covering Hoare logic (~100 students).

Informatica Introductieproject

BSc Informatica Year 1, Tutor, 21/22-24/25

Supervised students who worked in groups (2 or 3 groups of 5/6 students each) on developing an application (including, e.g., games and phone apps). Helped students by integrating techniques from Software Engineering to simplify the organization of the project and monitor its development.

Multi-agent Systems

MSc AI, Guest lecturer, 23/24

Gave a guest lecture on (multi-agent) reinforcement learning and reward machines (~40 students).

Theses

BSc AI, MSc AI

Supervised 13 BSc AI theses, mainly in RL, and 2 MSc AI theses. One of the two MSc AI theses is currently under submission at RLC 2026.

Educational Reading Club

Utrecht University

Took part to meetings of the educational reading club, discussing research papers on how to improve teaching.

Grants and Funding

Science For Sustainability Community Building Fund

Mar-Apr 2025

Secured funding to host for two weeks in Utrecht a visiting researcher on themes connecting AI to sustainability.

TAILOR Connectivity Fund

Jul 2022

Secured funding for a 1-month visit at a research group at Università degli Studi di Roma "La Sapienza". [C-5] was the main result of this visit.

Reviewing

- Conferences: AiML 2022, AAMAS 2024, ECAI 2024, AAMAS 2025, JELIA 2025, AAMAS 2026, ECIS 2026, NWO ICT.OPEN 2026
- Workshops: CMAS 2023, ALA 2026

Academic activities and outreach

Invited talk at the Research Seminar in Logic, Gothenburg University

May 2025

Invited to give a talk at the Research Seminar in Logic at Gothenburg University over my paper "Frame Definability in Conditional Logic" published at AiML 2024.

S4S Café - Sustainability for AI

Apr 2025

Co-organized a seminar on AI and sustainability, bringing together researchers from different fields to discuss topics on sustainability in AI.

VvL PhD day 2024

Jun 2024

Organized the 2024 edition of the VvL PhD day.

VvL Essentials

2023-2024

Organized a series of seminars on various research topics from researchers from several Dutch universities.